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LipScribe: Lip-Assisted Speech Recognition for Robust Live Captioning

COLE RODNEY MIGISHA[1], CHARIS OPOL[1], AINOMUGISHA FAITH[1], LISA LUSINGA NAMUGENYI[1], ANSASIRE NEHUM NAAMAYE[1], and WAKISA ISABELLE[1]

[1]Department of Computer Science, Uganda Christian University, Mukono, Uganda (e-mail: {b29657, b29654, b27518, b30300, b29785, b30311}@students.ucu.ac.ug)

This work is submitted as the written project proposal for the LipScribe project in the Computer Vision and Deep Learning course, Group 4.

ABSTRACT Automatic Speech Recognition systems fail when background noise corrupts the audio signal. Lip movements remain completely unaffected by acoustic noise and therefore provide a reliable complementary channel. LipScribe is a multimodal system that improves transcription by combining audio understanding with lip reading. The system crops the mouth region using MediaPipe Face Mesh, extracts spatiotemporal features with a 3D Convolutional Neural Network, processes audio with recurrent networks, and fuses both streams using attention that adapts to noise level. Training uses the GRID corpus with controlled noise injection. Evaluation is performed with Word Error Rate and Character Error Rate across multiple signal-to-noise ratios. The primary beneficiaries are deaf and hard-of-hearing users who depend on accurate live captions in lectures, meetings and public spaces.

INDEX TERMS Audiovisual speech recognition, lip reading, multimodal fusion, connectionist temporal classification, MediaPipe, live captioning, assistive technology, noise robustness

TABLE OF ABBREVIATIONS

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I. INTRODUCTION

A. BACKGROUND

Traditional Automatic Speech Recognition (ASR) systems work well in quiet rooms but degrade sharply when background noise, overlapping speakers or muffled speech corrupt the audio signal.[1] Lip movements, however, are completely unaffected by acoustic noise. By watching the mouth, a computer can still obtain useful information about what is being said. Combining the audio signal with visual lip information therefore offers a more reliable route to accurate live captions, especially in real-world noisy environments.

B. PROBLEM STATEMENT

Most current live-captioning tools rely only on audio. In noisy classrooms, public transport, industrial sites or outdoor events these tools produce high error rates. The central problem addressed by this project is:

> *How can we improve transcription by combining audio understanding with lip reading so that captions remain usable even when the audio channel is severely degraded?*

C. RESEARCH OBJECTIVES

The project has four concrete objectives:

1) **Disambiguate homophenes**: Learn to distinguish lip shapes that look almost identical but correspond to different letters or words (for example "mat" versus "bat").[2]
2) **Model temporal word sequences**: Process continuous sequences of video frames so that the system recognises whole words and short phrases rather than isolated letters.
3) **Detect emotion from speaking style**: Capture emotional information from the dynamics of lip and facial movement while the person is speaking.
4) **Identify the spoken language**: Recognise which language the speaker is using by jointly analysing lip movements and audio.

[1]Automatic Speech Recognition (ASR) means converting spoken words into written text.

[2]Homophenes are different sounds that produce almost the same lip shape.

TABLE 1. Table of Abbreviations

Abbreviation	Meaning
ASR	Automatic Speech Recognition 1
CER	Character Error Rate 3
CTC	Connectionist Temporal Classification 3
MFCC	Mel-Frequency Cepstral Coefficients 3
SNR	Signal-to-Noise Ratio 3
WER	Word Error Rate 3

D. JUSTIFICATION

More than 466 million people worldwide live with hearing loss and depend on captions or lip reading for everyday communication. Human sign-language interpreters are scarce and commercial assistive devices remain expensive. A practical system that improves live captions in noisy conditions can increase accessibility in education, healthcare and public services. From a technical perspective the project is well suited to a six-week computer-vision capstone: it requires real data handling, a clear baseline, transfer learning, and the integration of at least two advanced capabilities (spatiotemporal visual encoding and adaptive multimodal fusion).

II. LITERATURE REVIEW

A. MAIN RESEARCH QUESTION

How can a multimodal deep learning system effectively integrate visual lip movements with audio signals to produce accurate live captions in noisy environments?

B. SPECIFIC RESEARCH QUESTIONS

From the four project objectives we derive the following research questions:

1) **RQ1 (Homophene Disambiguation):** How can an audio-visual speech recognition system distinguish between words or letters that share visually identical lip movements?
2) **RQ2 (Temporal Sequence Modeling):** Which spatiotemporal deep learning architectures and loss functions best capture frame sequences for full-word and sentence transcription?
3) **RQ3 (Emotion Cues & Speaking Style):** How can real-time facial landmark detection and geometric mesh encodings extract emotion cues from facial expressions?
4) **RQ4 (Language Identification):** How can visual lip-reading models adapt to distinct spoken languages, accents, and regional scripts?

C. RESULTS FROM THE LITERATURE

1) RQ1: Disambiguating Homophenes

Homophenes are different sounds that produce almost identical lip shapes (for example /p/ versus /b/, or “pat” versus “bat”) [1]. A single video frame cannot resolve this ambiguity because one visual shape can correspond to several possible sounds [2]. Recent work addresses the problem by storing one-to-many mappings between visual features and candidate audio features inside multi-head Key-Value memory networks, and by using joint cross-attention so that the audio context can resolve the visual ambiguity [2]. Temporal modelling over neighbouring frames also helps separate similar-looking sounds [3].

2) RQ2: Modelling Temporal Word Sequences

Three-dimensional convolutional networks extract both the shape of the lips and their motion across consecutive frames at the same time [1]. When these 3D features (or 2D features from EfficientNetB0 / NASNetMobile) are passed to bidirectional LSTM layers, the network can learn forward and backward temporal dependencies [4], [5]. Connectionist Temporal Classification (CTC) loss is especially useful because it maps a variable-length sequence of frames directly to text without requiring expensive frame-level labels [4], [6].

3) RQ3: Emotion Detection from Facial Expressions

Facial expressions carry a large part of the emotional information that accompanies speech [7]. MediaPipe Face Mesh detects 468 three-dimensional facial landmarks in real time [3]. By selecting the landmarks that correspond to the Facial Action Coding System and converting the resulting mesh angles into compact feature vectors, researchers have reached emotion classification accuracies up to 97 % [7].

4) RQ4: Language Identification and Regional Scripts

Lip movements differ across languages, so models trained on one language do not automatically generalise to another. Dedicated datasets such as IndoLR (Indonesian) and the Kannada corpus used by KanAVNet have been created to address this issue [3], [4]. Dual-stream networks that map both acoustic features and visual lip embeddings into the target script, combined with attention-guided late fusion, have shown good results on regional languages and accents [4].

D. DISCUSSION

The literature consistently shows that pure audio systems fail in noise while visual lip features remain stable [2]. MediaPipe Face Mesh provides a fast and reliable way to isolate the mouth [3]. Spatiotemporal models that combine

3D convolutions (or efficient 2D backbones) with bidirectional LSTMs, trained with CTC loss, currently offer the best trade-off between accuracy and practicality [1], [4]. Attention-based fusion further improves robustness by giving more weight to the visual stream when the audio becomes unreliable [2].

LipScribe will directly apply these findings: MediaPipe for mouth cropping, a 3D-CNN + EfficientNetB0 visual stream paired with a BiLSTM, a parallel audio BiLSTM, cross-attention fusion that adapts to noise level, and CTC decoding with beam search.

E. FUTURE WORK AND REMAINING GAPS

Several important problems remain open. Models still struggle with large head pose changes and severe lighting variation [5]. Moving from small closed vocabularies to unconstrained natural speech is still difficult [6]. Occlusions caused by hands, facial hair or masks continue to degrade landmark detection [4]. Real-time performance on mobile devices and strong privacy guarantees for continuous camera use also require further research [3], [8].

III. METHODOLOGY

A. DATASET AND NOISE PREPARATION

The primary dataset is the GRID audiovisual sentence corpus, which contains 34 speakers and approximately 33 000 short sentences with clean frontal video and aligned audio. Synthetic noise from the MUSAN corpus will be mixed into the audio at controlled Signal-to-Noise Ratio (SNR) levels (clean, 10 dB, 5 dB, 0 dB and −5 dB).[3]

B. MOUTH CROPPING

MediaPipe Face Mesh will be applied to every video frame. The lip keypoints are used to compute a tight bounding box, which is then cropped and resized to a fixed resolution (96×96 or 112×112).

C. DUAL-STREAM ARCHITECTURE

- **Visual stream**: Cropped mouth sequences pass through a 3D convolutional front-end followed by a bidirectional LSTM.
- **Audio stream**: Raw audio is converted to log-Mel spectrograms and Mel-Frequency Cepstral Coefficients (MFCC) features, then encoded by a bidirectional LSTM.

D. MULTIMODAL FUSION

Audio and visual features are combined by an attention-based fusion module that automatically gives more weight to the lips when the audio is noisy. During training we also use modality dropout so the network cannot ignore the visual stream.

[3]Signal-to-Noise Ratio (SNR) tells us how strong the voice is compared with the background noise. Lower SNR means more noise.

E. BASELINES AND ADVANCED TECHNIQUES

We first train three simple baselines (audio-only, visual-only, and simple concatenation). The final system adds two advanced capabilities: (1) a 3D convolutional visual encoder and (2) adaptive cross-modal attention with modality dropout. Sequence prediction uses Connectionist Temporal Classification (CTC) loss with beam search.

F. EVALUATION

Performance is measured by Word Error Rate (WER) and Character Error Rate (CER) on held-out speakers at every tested SNR. Ablation studies will show the contribution of each advanced component.

IV. EXPECTED RESULTS

Under clean conditions the fused model is expected to achieve a Word Error Rate below 10 %. As noise increases, the visual stream should increasingly compensate for the degraded audio, producing a clear advantage over the audio-only baseline at 0 dB and −5 dB SNR. The final deliverable will include a clean inference script and a simple web demonstration.

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